What Motion Can You Detect?

SYNOPSIS

Before we can talk about forces, we have to settle something surprisingly slippery: which motions are real, and which only look real because of where you happen to be standing. The answer turns out to decide the shape of every law in this module — including why Newton's second law is about *acceleration* and not about speed.

BEFORE YOU READ ON — PREDICT

You are a passenger in a windowless van with no instruments, riding on a perfectly smooth, straight road. Two questions:

Can you tell, from inside, whether the van is **parked** or **cruising at 60 mph**? And can you tell whether the driver has just **hit the brakes**?

Commit to yes or no for each, and one sentence on how you would find out, before you read on.

PERFORMANCE EXPECTATIONS

1. **Decide**, for a described situation inside a closed vehicle, whether the motion could be detected from inside it — and say what experiment would settle it.
2. **State** the principle of relativity in your own words, and explain how it follows from inertia.
3. **Explain** why static and dynamic equilibrium are the same physical state, not two states.
4. **Explain** why Newton's second law relates force to acceleration rather than to velocity.
5. **Draw** a free-body diagram for a single body, and say what belongs on it and what does not.

Three experiments you have already run

1 · Drop a ball in the van

Hold a ball at shoulder height and let go. Where does it land — at your feet, or somewhere behind you, since the van moved forward while the ball was falling?

At your feet. Exactly as it would if the van were parked in a driveway.

2 · Coffee at 900 kilometres an hour

A cup of coffee on a tray table at cruising altitude sits perfectly still. The aircraft is moving faster than almost anything you will ever ride in, and the surface of the coffee does not know it. Then the pilot throttles back, or hits turbulence, or lands — and it slops over the rim.

Speed: undetectable. Change of speed: unmistakable.

3 · Standing on the subway

Watch the standing passengers. They lean back as the train pulls out, and forward as it brakes into the next station. In between — at forty miles an hour through a tunnel — they stand up straight and read their phones.

Nobody braces against speed. Everybody braces against a change in it.

What the three have in common

1. State the **Galilean principle of relativity** in your own words.

2. How is the principle of relativity connected to the idea of **inertia**?

3. State **Newton's first law**, and describe why it contains both of the ideas above.

THE CLAIM THIS NOTE RESTS ON

There is no experiment you can perform to decide which frame is *really* at rest. **Velocity cannot be measured absolutely — only relative to something else. Acceleration can.**

Which is why Newton's second law is written $F = ma$ and not $F = mv$. A law built on velocity would give different answers to different observers, and would not be a law at all. A law built on acceleration gives the same answer to everyone — so acceleration is the quantity a force can be held responsible for.

WITH CALCULUS

Everything above is an argument from examples. It is also a derivation, and the derivation is two lines long.

Let frame S' move along x at a constant velocity V relative to frame S. The two observers put the same event at

$$x' = x - Vt$$

1. Differentiate once with respect to t. Write v' in terms of v.

2. Differentiate again. Write a'.

3. Which of those two steps made V disappear — and what property of V was responsible?

4. Now let the second frame **accelerate**, so that $x' = x - \frac{1}{2}At^2$. What is a' this time? What does your answer say about which frames the principle of relativity actually covers?

Does it hold? Four consequences to check

1. What is a **static equilibrium**? Give an example.

2. Can many forces act on an object that is in static equilibrium? Explain with a diagram.

3. "An object moving at constant velocity has zero net force, so the forces are in equilibrium." True or false, and why?

4. What is a **dynamic equilibrium**, how does it differ from a static one — and **what is the same in both**?

THE TOOL FOR THE REST OF THE MODULE

The free-body diagram

One body. Every force acting *on* it. Nothing else — not the forces it exerts on other things, not the forces acting on its surroundings, not the motion it happens to have.

That restriction sounds fussy now and will save you repeatedly later. It is the reason a force and its Newton's-third-law partner can never appear on the same diagram: they act on two different bodies, so they belong to two different diagrams.

1. What is a **free-body diagram**? Give two examples.

BACK TO YOUR PREDICTION

You predicted whether you could tell, from inside the windowless van, that it was moving — and whether you could tell it had braked. Which of your two answers was right, and which of the three experiments settled it?

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